

## Title

Feasibility and Efficacy of Wearable Robotic Assistance for Motor Rehabilitation in Parkinson Disease: A Pilot Randomized Controlled Study Protocol

## Abstract

Motor dysfunction and gait impairment in Parkinson disease significantly restrict mobility and independence. Emerging robotic technologies may augment traditional physical rehabilitation by providing consistent, quantifiable assistance. This pilot protocol evaluates whether supplementing conventional physical therapy with structured sessions in a soft robotic exoskeleton improves locomotor function and motor symptoms in patients with Parkinson disease. Approximately 56 individuals with Parkinson disease will be randomly allocated to either conventional rehabilitation (control) or identical conventional therapy augmented with soft exoskeleton-assisted training. All participants will receive five sessions weekly for four weeks. Primary outcomes include gait velocity and stride symmetry. Secondary measures encompass ankle range of motion, motor severity scales, and cognitive assessment. We hypothesize that robotic assistance during training facilitates greater locomotor gains than conventional therapy alone.